

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 05 · Constant Acceleration in 2D

With Answers

Questions with full worked solutions

- 10 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 05

Constant Acceleration in 2D

Your classified route

4

QUESTIONS

38

MARKS

1

FAMILIES

3

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 05 | FAMILY 01

Constant-acceleration vector kinematics

WHAT THIS FAMILY ASKS

Use componentwise displacement, velocity and acceleration relationships for motion in a plane.

VARIANT ROUTE

V01 Recover constant acceleration

V02 Update velocity with $v = u + at$

V03 Two accelerated particles with equal speed/direction conditions

4 classified questions

38 total marks

Summer 2022 Q6 — complete original question

6. A particle P is moving with constant acceleration.

At time $t = 1$ second, P has velocity $(-\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$

At time $t = 4$ seconds, P has velocity $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

Find the speed of P at time $t = 3.5$ seconds.

(6)

DR ESLAM AHMED

Summer 2022 Q6 — detailed solution

Recover constant acceleration

ALL
6 marks**Find the acceleration**

The velocity changes from $-\mathbf{i} + 4\mathbf{j}$ at $t = 1$ to $5\mathbf{i} - 8\mathbf{j}$ at $t = 4$. The time difference is 3 s.

Calculate acceleration

$$\mathbf{a} = \frac{(5\mathbf{i} - 8\mathbf{j}) - (-\mathbf{i} + 4\mathbf{j})}{3} = \frac{6\mathbf{i} - 12\mathbf{j}}{3} = 2\mathbf{i} - 4\mathbf{j}.$$

Find velocity at $t = 3.5$

From $t = 1$ to $t = 3.5$ is 2.5 s, so $\mathbf{v} = (-\mathbf{i} + 4\mathbf{j}) + 2.5(2\mathbf{i} - 4\mathbf{j})$.

Simplify the velocity

$$\mathbf{v} = -\mathbf{i} + 4\mathbf{j} + 5\mathbf{i} - 10\mathbf{j} = 4\mathbf{i} - 6\mathbf{j}.$$

Find the speed

$$\text{Speed} = \sqrt{4^2 + (-6)^2} = \sqrt{52} = 2\sqrt{13} \text{ m s}^{-1}, \text{ about } 7.2 \text{ m s}^{-1}.$$

Final answer speed = $\sqrt{52} = 2\sqrt{13} \text{ m s}^{-1} \approx 7.2 \text{ m s}^{-1}$.

Summer 2021 Q5 — complete original question

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

A particle P is moving with constant acceleration. At 2 pm, the velocity of P is $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$ and at 2.30 pm the velocity of P is $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time T hours after 2 pm, P is moving in the direction of the vector $(-\mathbf{i} + 2\mathbf{j})$

- (a) Find the value of T .

(6)

Another particle, Q , has velocity $\mathbf{v}_Q \text{ km h}^{-1}$ at time t hours after 2 pm, where

$$\mathbf{v}_Q = (-4 - 2t)\mathbf{i} + (\mu + 3t)\mathbf{j}$$

and μ is a constant.

Given that there is an instant when the velocity of P is equal to the velocity of Q ,

- (b) find the value of μ .

(3)

Summer 2021 Q5 — detailed solution**ALL**
9 marks**Find P acceleration**

From 2:00 to 2:30 is 0.5 hours. The velocity changes from $3\mathbf{i} + 5\mathbf{j}$ to $\mathbf{i} + 7\mathbf{j}$.

Calculate acceleration

$$\mathbf{a} = \frac{(\mathbf{i} + 7\mathbf{j}) - (3\mathbf{i} + 5\mathbf{j})}{0.5} = -4\mathbf{i} + 4\mathbf{j}.$$

Write P velocity

At time t hours after 2:00, $\mathbf{v}_P = (3 - 4t)\mathbf{i} + (5 + 4t)\mathbf{j}$.

Use the direction vector

At time T , the velocity is parallel to $-\mathbf{i} + 2\mathbf{j}$, so $\frac{5 + 4T}{3 - 4T} = \frac{2}{-1} = -2$.

Find T

$5 + 4T = -2(3 - 4T) = -6 + 8T$. Hence $11 = 4T$ and $T = \frac{11}{4}$ hours.

Equate i-components for P and Q

For equal velocities, $3 - 4t = -4 - 2t$. This gives $7 = 2t$, so $t = 3.5$.

Equate j-components

At $t = 3.5$, P's j-component is $5 + 4(3.5) = 19$. Q's j-component is $\mu + 3(3.5) = \mu + 10.5$.

Find mu

$\mu + 10.5 = 19$, so $\mu = 8.5$.

Final answer (a) $T = \frac{11}{4}$ h. (b) $\mu = 8.5$.

October 2023 Q4 — complete original question

4. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively.]

A particle P moves with constant acceleration $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})\text{ m s}^{-2}$, where λ is a positive constant.

At time $t = 0$, the velocity of P is $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

- (a) Find the velocity of P when $t = 5\text{ s}$, giving your answer in terms of $\mathbf{i}$, $\mathbf{j}$ and λ . (2)

The speed of P when $t = 5\text{ s}$ is 13 m s^{-1}

- (b) Show that

$$25\lambda^2 - 42\lambda - 16 = 0 \quad (3)$$

- (c) Find the direction of motion of P when $t = 4\text{ s}$, giving your answer as a bearing to the nearest degree. (5)

October 2023 Q4 — detailed solutionUpdate velocity with $\mathbf{v} = \mathbf{u} + \mathbf{at}$ **ALL**
10 marks**Use $\mathbf{v} = \mathbf{u} + \mathbf{at}$** The initial velocity is $(5\mathbf{i} - 8\mathbf{j})$ and the acceleration is $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})$. At $t = 5$, $\mathbf{v} = (5\mathbf{i} - 8\mathbf{j}) + 5(-\lambda\mathbf{i} + 2\lambda\mathbf{j})$.**Write the velocity at $t = 5$** So $\mathbf{v} = (5 - 5\lambda)\mathbf{i} + (-8 + 10\lambda)\mathbf{j}$.**Use the speed at $t = 5$** The speed is 13, so $(5 - 5\lambda)^2 + (-8 + 10\lambda)^2 = 13^2$.**Simplify to the printed equation**Expanding gives $25 - 50\lambda + 25\lambda^2 + 64 - 160\lambda + 100\lambda^2 = 169$. Therefore $125\lambda^2 - 210\lambda - 80 = 0$, and dividing by 5 gives $25\lambda^2 - 42\lambda - 16 = 0$.**Find lambda**Solving $25\lambda^2 - 42\lambda - 16 = 0$ gives $\lambda = 2$ or $\lambda = -\frac{8}{25}$. Since λ is positive, $\lambda = 2$.**Find the velocity at $t = 4$** $\mathbf{v}(4) = (5\mathbf{i} - 8\mathbf{j}) + 4(-2\mathbf{i} + 4\mathbf{j}) = -3\mathbf{i} + 8\mathbf{j}$.**Convert to a bearing**The motion is north-west. The angle west of north satisfies $\tan \alpha = \frac{3}{8}$, so $\alpha = 20.6^\circ$. The bearing is $360^\circ - 20.6^\circ = 339.4^\circ$, so 339° to the nearest degree.**Final answer** (a) $\mathbf{v} = (5 - 5\lambda)\mathbf{i} + (-8 + 10\lambda)\mathbf{j}$. (b) $25\lambda^2 - 42\lambda - 16 = 0$. (c) bearing = 339° .

Summer 2025 Q7 — complete original question

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

Two boats, A and B , are each moving with constant acceleration.

At time t hours after noon, boat A has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

- (a) Find the magnitude of the acceleration of A . (2)

When $t = 2$, the velocity of B is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When $t = 5$, the velocity of B is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

- (b) Find the acceleration of B , giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)

- (c) Find the velocity of B at time $t = 0$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)

At time T_1 hours after noon, both boats are moving with the same speed.

- (d) Find the exact value of T_1 (4)

At time T_2 hours after noon, both boats are moving in the same direction.

- (e) Show that $3T_2^2 + pT_2 + q = 0$, where p and q are integers to be found. (3)

Summer 2025 Q7 — detailed solution

Equal-speed and equal-direction conditions

ALL
13 marks**Part (a): Read A's acceleration**From $\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$, the constant acceleration is $\mathbf{a}_A = \mathbf{i} - 4\mathbf{j} \text{ km h}^{-2}$.**Part (a): Find its magnitude** $|\mathbf{a}_A| = \sqrt{1^2 + (-4)^2} = \sqrt{17} = 4.1 \text{ km h}^{-2}$ to 2 significant figures.**Part (b): Recover B's acceleration**

$$\mathbf{a}_B = \frac{(\mathbf{i} - 5\mathbf{j}) - (4\mathbf{i} + \mathbf{j})}{5 - 2} = -\mathbf{i} - 2\mathbf{j} \text{ km h}^{-2}.$$

Part (c): Work backwards to $t = 0$

$$\mathbf{v}_B(0) = \mathbf{v}_B(2) + \mathbf{a}_B(0 - 2) = (4\mathbf{i} + \mathbf{j}) + (-\mathbf{i} - 2\mathbf{j})(-2) = 6\mathbf{i} + 5\mathbf{j}.$$

Part (d): Write both velocity vectors

$$\mathbf{v}_A = (2 + t)\mathbf{i} + (3 - 4t)\mathbf{j} \text{ and } \mathbf{v}_B = (6 - t)\mathbf{i} + (5 - 2t)\mathbf{j}.$$

Part (d): Equate the squared speeds $|\mathbf{v}_A|^2 = 17t^2 - 20t + 13$ and $|\mathbf{v}_B|^2 = 5t^2 - 32t + 61$. Hence $12t^2 + 12t - 48 = 0$, so $t^2 + t - 4 = 0$.**Part (d): Choose the positive time**

$$t = \frac{-1 \pm \sqrt{17}}{2}, \text{ so } T_1 = \frac{-1 + \sqrt{17}}{2} \text{ hours.}$$

Part (e): Use the same-direction conditionParallel velocity vectors give $\frac{2 + T_2}{6 - T_2} = \frac{3 - 4T_2}{5 - 2T_2}$. Cross-multiplying gives $(2 + T_2)(5 - 2T_2) = (3 - 4T_2)(6 - T_2)$.**Part (e): Match the required quadratic**Expansion gives $6T_2^2 - 28T_2 + 8 = 0$, hence $3T_2^2 - 14T_2 + 4 = 0$. Therefore $p = -14$ and $q = 4$.**Final answer** (a) $|\mathbf{a}_A| = \sqrt{17} \text{ km h}^{-2} \approx 4.1 \text{ km h}^{-2}$. (b) $\mathbf{a}_B = -\mathbf{i} - 2\mathbf{j} \text{ km h}^{-2}$. (c) $\mathbf{v}_B(0) = 6\mathbf{i} + 5\mathbf{j} \text{ km h}^{-1}$. (d) $T_1 = \frac{-1 + \sqrt{17}}{2} \text{ hours}$. (e) $3T_2^2 - 14T_2 + 4 = 0$, $p = -14$, $q = 4$.